CAMBRIDGE IGCSE & O LEVEL COMPUTER SCIENCE (2210 / 0478)

Topic 2.1: Types and Methods of Data Transmission

COMPREHENSIVE REVISION GUIDE & STUDY NOTES

1. DATA PACKETS AND PACKET SWITCHING (2.1.1)

What is a Data Packet?

When data is transmitted across a network or the Internet, large files are broken down into smaller manageable chunks called **data packets**. This prevents network congestion and allows reliable transmission.

Structure of a Data Packet (3 Core Parts):

Packet Component	Information Contained	Purpose
1. Packet Header	• Sender's IP address (Source IP) • Receiver's IP address (Destination IP) • Packet sequence number (e.g., Packet 3 of 10) • Packet size (in bytes)	Ensures the packet reaches the correct destination and can be reassembled in the right order.
2. Payload	• The actual chunk of user data being transmitted (e.g., part of a web page, image, or email).	Delivers the actual content (usually 1 KB to 1.5 KB per packet).
3. Packet Trailer	• End-of-packet marker (indicates the packet is complete) • Error detection data (Checksum or Cyclic Redundancy Check - CRC)	Verifies whether the packet arrived intact without transmission errors.

How Packet Switching Works:

1. The original file is split into numbered packets.
2. Each packet travels **independently** through the network via **routers**.
3. Packets may take **different routes** depending on network traffic, line availability, and speed.
4. Packets may arrive **out of order** at the receiving computer.
5. The receiving computer checks for errors using the trailer checksum and uses sequence numbers to **reassemble** them into the original file.
6. If a packet is corrupted or missing, a request is sent to retransmit only that specific packet.

2. DIRECTION OF DATA TRANSMISSION (2.1.2)

Data transmission can occur in three different directional modes:

Direction Mode	Definition & Working	Real-World Examples
Simplex	Data travels in **ONE direction only** (unidirectional). The receiver cannot send data back to the sender over the same channel.	• Keyboard to CPU • Computer to Monitor / Printer • Microphone to Speaker
Half-Duplex	Data travels in **BOTH directions**, but **NOT at the same time** (one direction at a time). While one transmits, the other listens.	• Walkie-talkies (Push-to-Talk) • CB radio communication • Some industrial sensor channels
Full-Duplex	Data travels in **BOTH directions SIMULTANEOUSLY** (at the same time) over separate channels or duplex lines.	• Telephone / Mobile calls • Broadband Internet connection • Video conferencing (Zoom/Teams)

3. METHODS OF DATA TRANSMISSION: SERIAL VS. PARALLEL (2.1.3)

Data can be transmitted using either a single transmission line (Serial) or multiple lines (Parallel):

Feature	Serial Data Transmission	Parallel Data Transmission
How bits are sent	Bits are sent **one bit at a time** down a single wire or channel.	Multiple bits (e.g., 8 bits / 1 byte) are sent **simultaneously** down multiple wires.
Distance suitability	Ideal for **long distances** (kilometers, Internet, networks).	Suitable only for **short distances** (typically under a few meters, internal buses).
Bit Skew risk	**No bit skew** (bits follow each other in order on a single line).	**High risk of bit skew** (bits travel at slightly different speeds on different wires and arrive out of sync).
Crosstalk / Interference	Very low electrical interference between wires.	High risk of **crosstalk** (electrical signals bleeding into adjacent parallel wires).
Hardware Cost	Cheaper (requires fewer wires and connectors).	More expensive (requires multi-core cables and complex connectors).
Common Uses	USB cables, Ethernet, Fibre optic cables, SATA.	Internal computer buses (Data Bus), older ribbon cables (IDE, Centronics printer port).

⚠️ **KEY EXAM DEFINITIONS:**

- **Bit Skew:** When bits transmitted simultaneously across parallel wires arrive at the destination at slightly different times due to minor physical variations in the wires.
- **Crosstalk:** Electromagnetic interference between adjacent wires in a parallel cable causing data corruption.

4. UNIVERSAL SERIAL BUS (USB) TECHNOLOGY (2.1.4)

What is USB?

USB (Universal Serial Bus) is an asynchronous serial data transmission method designed to connect peripheral devices to computers.

How USB Works:

- Standard USB cables contain **4 wires**:
  - **2 wires for power:** +5V power supply and Ground (0V).
  - **2 wires for data:** Data+ and Data- (uses differential signaling to cancel noise).
- When a USB device is plugged in, the computer automatically detects the voltage change and queries the device.
- The OS recognizes the device, loads the appropriate **device driver**, and assigns it an address.

Key Advantages and Disadvantages of USB:

Advantages of USB	Disadvantages / Limitations of USB
• **Plug and Play:** Devices are automatically detected and configured. • **Provides Power:** Eliminates the need for external power supplies for mice, keyboards, and flash drives. • **Backward Compatible:** Newer USB ports support older USB version devices. • **Universal Standard:** Prevents incorrect port connections and replaces multiple older legacy ports (serial, parallel, PS/2). • **High Data Transfer Rates:** Modern USB standards (USB 3.0/Type-C) support very fast transfer speeds.	• **Cable Length Limited:** Standard USB cables are limited to a maximum length of about **5 meters** without an active booster. • **Transmission Speed vs Internal Buses:** While fast, USB is still slower than dedicated internal buses like PCIe. • **Older USB Standards:** USB 2.0 (max 480 Mbps) is slow compared to modern Gigabit network transmission.